

Cost of Capital and Capital Structure

Corporate Finance Class - PE FEB UI

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Learning Objectives

- Calculate WACC using DDM and SML approaches.
- Understand the after-tax cost of debt and preferred stock.
- Perform EPS vs. EBIT analysis for financial leverage.
- Explain Static Trade-off and Pecking-Order theories.

1. Estimating Cost of Equity (R_e)

There are two primary approaches to estimate the return required by shareholders:

Dividend Discount Model (DDM)

Focuses on future dividends.

$$R_e = \frac{D_1}{P_0} + g$$

Limitation: Only applicable for companies paying stable dividends.

Security Market Line (SML/CAPM)

Focuses on systematic risk (β).

$$R_e = R_f + \beta(E(R_m) - R_f)$$

Advantage: Adjusts for risk; applicable to non-dividend paying firms.

1. Estimating the Cost of Equity (R_e)

A. Dividend Discount Model (DDM)

$$R_e = \frac{D_1}{P_0} + g$$

Example: $P_0 = \$40$, $D_1 = \$2.50$, $g = 4\%$.

$$R_e = (2.50/40) + 0.04 = 10.25\%$$

B. SML / CAPM Approach

$$R_e = R_f + \beta(E(R_m) - R_f)$$

Example: $R_f = 3\%$, $\beta = 1.1$, Market Premium = 6%.

$$R_e = 3\% + 1.1(6\%) = 9.6\%$$

2. Cost of Debt and Preferred Stock

Cost of Debt (R_d)

- Use the **Yield to Maturity (YTM)**, not the coupon rate.
- Interest is tax-deductible: **After-tax cost:** $R_d \times (1 - T_c)$.
- *Illustration:* YTM is 8%, Tax rate is 25%.
Effective cost = $8\% \times (1 - 0.25) = 6\%$.

Cost of Preferred Stock (R_p)

- Preferred stock is a perpetuity and NOT tax-deductible..

$$R_p = \frac{D}{P_0}$$

- *Illustration:* Dividend is \$5, Price is \$95. $R_p = 5.26\%$.

3. Weighted Average Cost of Capital (WACC)

The WACC is the minimum return a company must earn to satisfy its creditors and owners.

$$WACC = w_e R_e + w_p R_p + w_d R_d (1 - T_c)$$

Case Study: Company X has the following capital structure:

Source	Market Value	Weight	Component Cost
Equity	\$6,000,000	60%	$R_e = 12\%$
Debt	\$3,000,000	30%	$R_d = 8\%$ (Pre-tax)
Pref. Stock	\$1,000,000	10%	$R_p = 9\%$

Tax Rate (T_c): 25%

Calculation

$$\begin{aligned} WACC &= (w_e \times R_e) + (w_p \times R_p) + (w_d \times R_d \times [1 - T_c]) \\ WACC &= (0.60 \times 0.12) + (0.10 \times 0.09) + (0.30 \times 0.08 \times 0.75) \\ WACC &= 0.072 + 0.009 + 0.018 = \mathbf{9.9\%} \end{aligned}$$

4. Financial Leverage: EPS vs. EBIT Analysis

Financial leverage involves substituting equity with debt. Financial leverage magnifies both gains and losses.

- **Upside:** Magnifies EPS when EBIT is high.
- **Downside:** Increases financial risk and the probability of bankruptcy.

Break-even EBIT

The EBIT level where $EPS_{Debt} = EPS_{NoDebt}$.

Break-even EBIT Formula:

$$\frac{(EBIT - I_1)(1 - T_c)}{N_1} = \frac{(EBIT - I_2)(1 - T_c)}{N_2}$$

Where I = Interest expense and N = Number of shares.

Illustration: EPS-EBIT Comparison

Company Y needs \$1M capital.

- **Plan A (All-Equity):** 20,000 shares at \$50.
- **Plan B (Levered):** \$500k debt at 10% interest, 10,000 shares.

Analysis:

Metric	EBIT = \$40,000	EBIT = \$100,000
EPS Plan A	\$2.00	\$5.00
EPS Plan B	$\$(-10\text{k})/10\text{k} = -\1.00	$(\$50\text{k})/10\text{k} = \5.00

Conclusion: At EBIT \$100k, Plan B (Levered) is superior. The break-even EBIT in this case is exactly \$100,000.

5. Capital Structure Theories

Static Trade-off Theory

- Firms balance the **Tax Shield** of debt against the **Costs of Financial Distress**.
- Optimal structure is where the benefit of debt is offset by bankruptcy risk.

Pecking-Order Theory

- Firms have a hierarchy of preference for financing:
 1. Internal Funds (Retained Earnings)
 2. Debt
 3. Equity (Last resort due to adverse selection/signaling).

5. Static Trade-off Theory

Firms borrow up to the point where the **marginal benefit of tax shield** equals the **marginal cost of financial distress**.

$$V_L = V_U + PV(\text{Tax Shield}) - PV(\text{Financial Distress Costs})$$

- **Tax Shield:** $T_c \times \text{Debt Amount}$.
- **Distress Costs:** Direct (Legal fees) and Indirect (Loss of customers, high employee turnover).
- **Optimal Structure:** The peak of the firm value curve.

6. Pecking Order Theory

Based on **Asymmetric Information** (Managers know more than investors). Firms avoid issuing equity because it may signal that the stock is overvalued.

The Hierarchy (The Pecking Order):

- 1 **Internal Funds:** Use retained earnings first (no signaling cost).
- 2 **Debt:** If internal funds are exhausted, issue debt (lower information cost).
- 3 **Equity:** Last resort (highest information asymmetry and negative signaling).

Insight: Profitable firms use less debt because they have enough internal funds.

Summary

- **WACC** is the minimum acceptable return for a project.
- **Financial Leverage** is a double-edged sword (Risk vs. Return).
- **Static Trade-off** looks for an "Optimal Balance."
- **Pecking Order** focuses on "Information and Signaling."

Thank You!

Questions & Answers